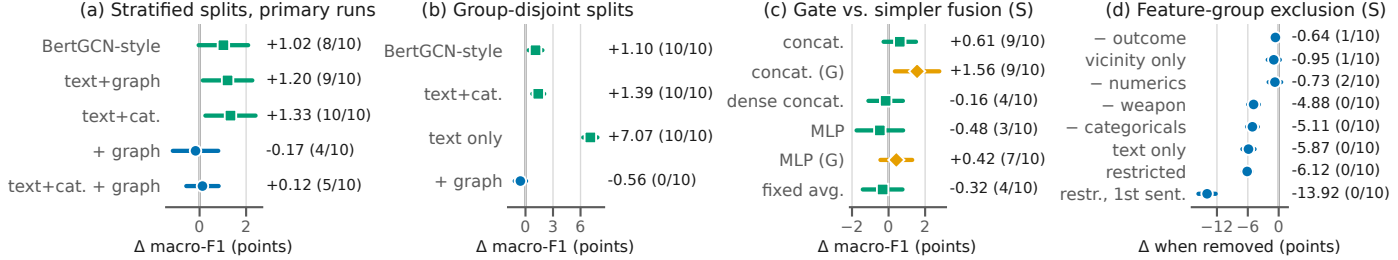
(a) Stratified splits, primary runs
BertGCN-style
text+graph
text+cat.
+ graph
text+cat. + graph
+1.02 (8/10)
+1.20 (9/10)
+1.33 (10/10)
-0.17 (4/10)
+0.12 (5/10)
0
2
Δ macro-F1 (points)
(b) Group-disjoint splits
BertGCN-style
text+cat.
text only
+ graph
+1.10 (10/10)
+1.39 (10/10)
+7.07 (10/10)
-0.56 (0/10)
0
3
6
Δ macro-F1 (points)
(c) Gate vs. simpler fusion (S)
concat.
concat. (G)
dense concat.
MLP
MLP (G)
fixed avg.
+0.61 (9/10)
+1.56 (9/10)
-0.16 (4/10)
-0.48 (3/10)
+0.42 (7/10)
-0.32 (4/10)
−2
0
2
Δ macro-F1 (points)
(d) Feature-group exclusion (S)
− outcome
vicinity only
− numerics
− weapon
− categoricals
text only
restricted
restr., 1st sent.
-0.64 (1/10)
-0.95 (1/10)
-0.73 (2/10)
-4.88 (0/10)
-5.11 (0/10)
-5.87 (0/10)
-6.12 (0/10)
-13.92 (0/10)
−12
−6
0
Δ when removed (points)